



IT Simplera Blue Team Internship

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Report : Week 1

❖ Wazuh SIEM Deployment and Configuration Report

• Introduction

Wazuh is an open-source Security Information and Event Management (SIEM) platform used for centralized security monitoring, log analysis, threat detection, and endpoint protection. In this internship task, the Wazuh Virtual Appliance was deployed using Oracle VirtualBox to act as the SIEM server. Two endpoints, Windows 11 and Kali Linux, were connected to the Wazuh Manager using Wazuh Agents. After successful agent registration, File Integrity Monitoring (FIM) was configured and tested by monitoring changes made to a test file. The generated security alerts were then verified through the Wazuh Dashboard to confirm that the monitoring environment was functioning correctly.

• Software Used

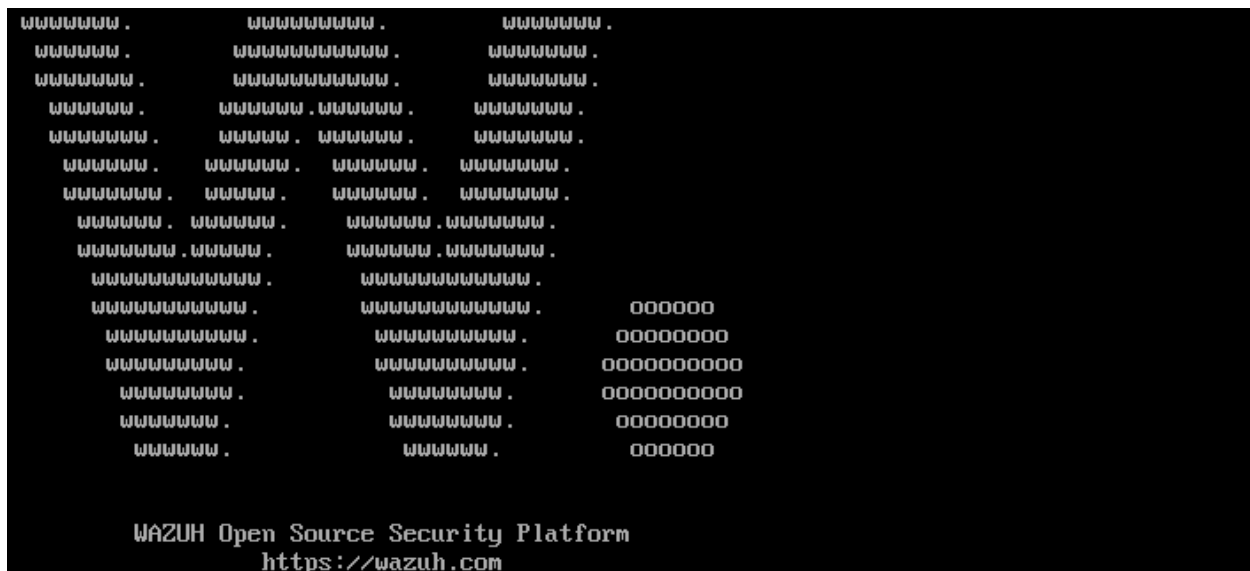
Software	Purpose
Oracle VirtualBox	Virtualization Platform
Wazuh OVA	SIEM Server
Kali Linux	Linux Endpoint
Windows 11	Windows Endpoint
Wazuh Agent 4.14.6-1	Endpoint Monitoring Agent

- **Deploying the Wazuh Virtual Appliance**

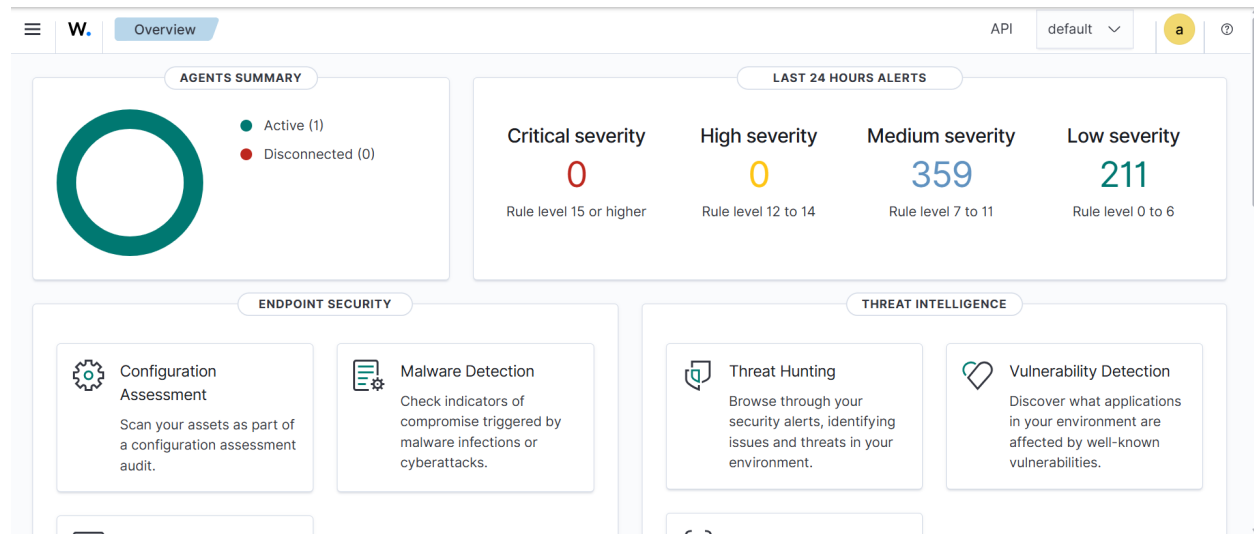
Procedure

The Wazuh Virtual Appliance (OVA) was imported into Oracle VirtualBox and powered on successfully. After the virtual machine booted, the Wazuh server terminal was accessed using the default credentials. The appliance provides the Wazuh Manager, Indexer, and Dashboard in a pre-configured environment, eliminating the need for a manual installation.

Wazuh Virtual Appliance Successfully Booted



Wazuh Dashboard Successfully Loaded



• Configuring the Wazuh Manager

The status of the Wazuh services was verified using the `systemctl` command. Initially, the Wazuh Manager service failed to start automatically after booting. The service was started manually using the `systemctl start wazuh-manager` command. After verification, the Wazuh Manager, Indexer, and Dashboard services were all running successfully.

Wazuh Manager Service Running

```

altname enp0s17
inet 10.06.27.132/24 metric 1024 brd 10.06.27.255 scope global dynamic eth0
    valid_lft 2187sec preferred_lft 2187sec
inet6 fe80::a00:27ff:fe07:531f/64 scope global dynamic mngtmpaddr noprefixroute
    valid_lft 6698sec preferred_lft 6698sec
inet6 fe80::a00:27ff:fe07:531f/64 scope link proto kernel_ll
    valid_lft forever preferred_lft forever
wazuh-user@wazuh-server ~$ sudo /var/ossec/bin/agent-control -lc
sudo: /var/ossec/bin/agent-control: command not found
wazuh-user@wazuh-server ~$ sudo /var/ossec/bin/agent-control -lc

Wazuh agent_control. List of available agents:
ID: 000, Name: wazuh-server (server), IP: 127.0.0.1, Active/Local
ID: 002, Name: Ahmad, IP: any, Active

wazuh-user@wazuh-server ~$ sudo systemctl status wazuh-manager
● wazuh-manager.service - Wazuh manager
   Loaded: loaded (/usr/lib/systemd/system/wazuh-manager.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-07-04 08:59:02 UTC; 1h 0min ago
     Process: 2621 ExecStart=/usr/bin/env /var/ossec/bin/wazuh-control start (code=exited, status=0/SUCCESS)
    Tasks: 232 (limit: 9477)
   Memory: 908.9M
         CPU: 7min 59.220s
   CGroup: /system.slice/wazuh-manager.service
           └─2694 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
             └─2695 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
               └─2696 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                 └─2699 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                   └─2702 /var/ossec/framework/python/bin/python3 /var/ossec/api/scripts/wazuh_apid.py
                     └─2740 /var/ossec/bin/wazuh-apid
                       └─2768 /var/ossec/bin/wazuh-db
                         └─2809 /var/ossec/bin/wazuh-execd
                           └─2820 /var/ossec/bin/wazuh-analyd
                             └─2850 /var/ossec/bin/wazuh-syscheckd
                               └─2871 /var/ossec/bin/wazuh-remoted
                                 └─2862 /var/ossec/bin/wazuh-logcollector
                                   └─2863 /var/ossec/bin/wazuh-monitord
                                     └─3000 /var/ossec/bin/wazuh-modulesd

Jul 04 08:49:58 wazuh-server eno[2621]: wazuh-logcollector: Process 2955 not used by Wazuh, removing...
Jul 04 08:49:58 wazuh-server eno[2621]: wazuh-monitord: Process 2976 not used by Wazuh, removing...
Jul 04 08:49:59 wazuh-server eno[2621]: Started wazuh-monitord...
Jul 04 08:49:59 wazuh-server eno[2621]: wazuh-modulesd: Process 2996 not used by Wazuh, removing...
Jul 04 08:49:59 wazuh-server eno[2981]: 2026-07-04 08:49:59 wazuh-modulesd:router: INFO: Loaded router module.
Jul 04 08:49:59 wazuh-server eno[2981]: 2026-07-04 08:49:59 wazuh-modulesd:content-manager: INFO: Loaded content manager module.
Jul 04 08:49:59 wazuh-server eno[2981]: 2026-07-04 08:49:59 wazuh-modulesd:inventory-harvester: INFO: Loaded inventory harvester module.
Jul 04 08:50:00 wazuh-server eno[2621]: Started wazuh-modulesd...
Jul 04 08:50:02 wazuh-server eno[2621]: Completed.
wazuh-user@wazuh-server ~$

```

Command Used

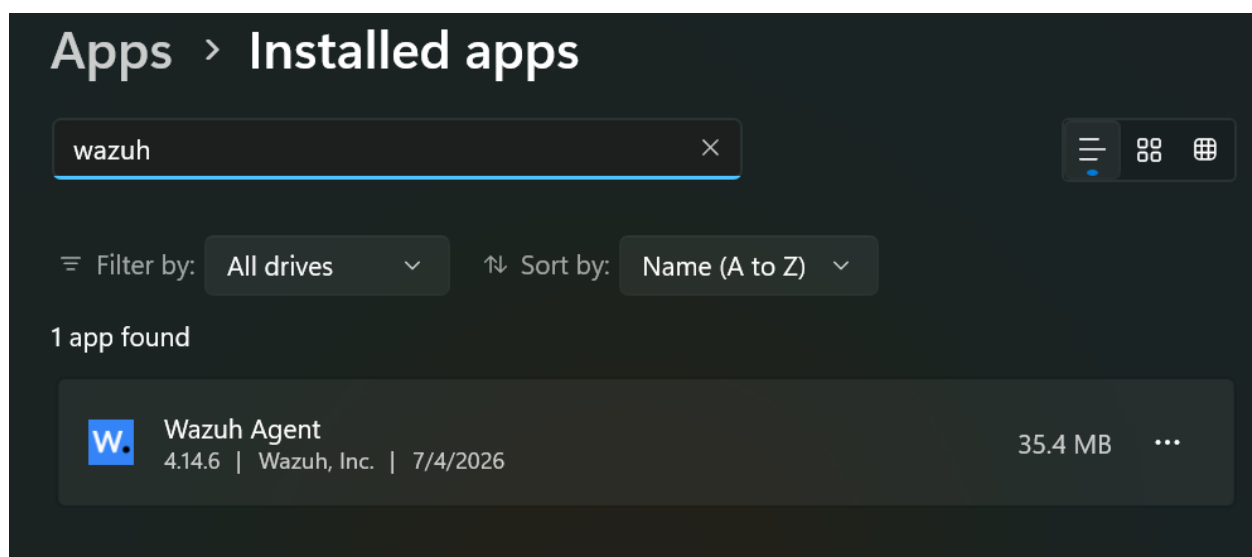
```
sudo systemctl status wazuh-manager  
sudo systemctl start wazuh-manager
```

• Installing the Windows Wazuh Agent

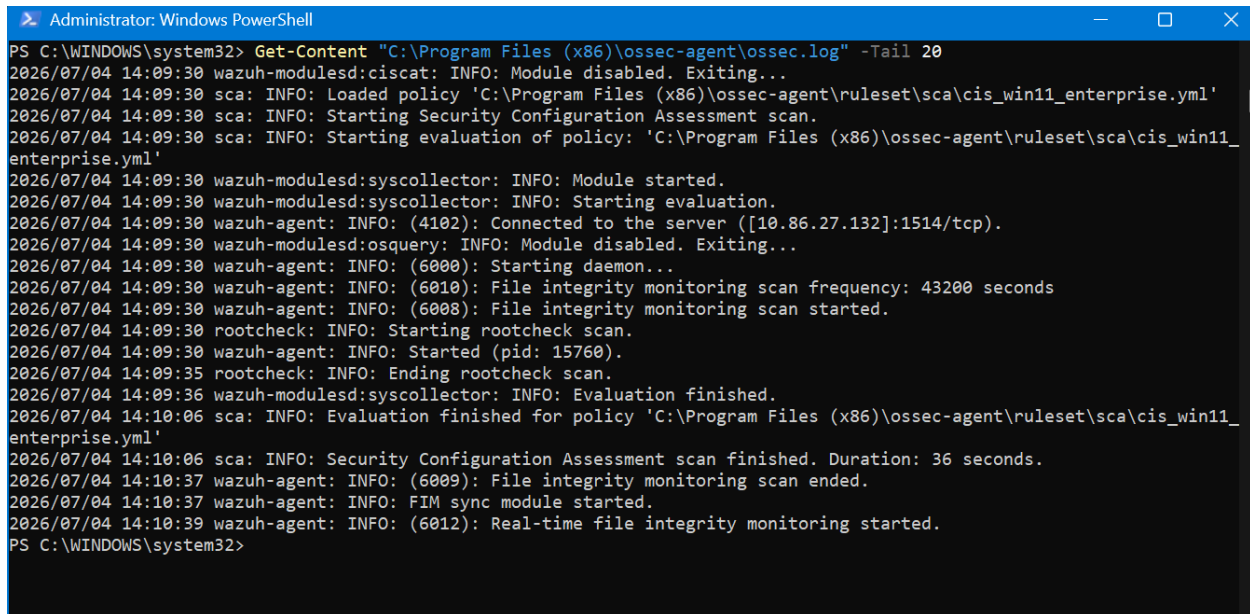
The official Wazuh Windows Agent installer (**version 4.14.6**) was downloaded from the Wazuh website and installed successfully. Since the installer did not request the manager IP address during installation, the agent configuration file (`ossec.conf`) was manually updated to include the correct Wazuh Manager IP address.

After modifying the configuration, the Wazuh Agent service was restarted.

Windows Wazuh Agent Installation



Windows Agent Successfully Connected

A screenshot of a Windows PowerShell window titled "Administrator: Windows PowerShell". The command prompt shows the execution of "Get-Content 'C:\Program Files (x86)\ossec-agent\ossec.log' -Tail 20". The output displays a series of log messages from the Wazuh agent, including module status updates, policy loading, and a successful connection to the server at [10.86.27.132]:1514/tcp.

```
PS C:\WINDOWS\system32> Get-Content "C:\Program Files (x86)\ossec-agent\ossec.log" -Tail 20
2026/07/04 14:09:30 wazuh-modulesd:ciscat: INFO: Module disabled. Exiting...
2026/07/04 14:09:30 sca: INFO: Loaded policy 'C:\Program Files (x86)\ossec-agent\ruleset\sca\cis_win11_enterprise.yml'
2026/07/04 14:09:30 sca: INFO: Starting Security Configuration Assessment scan.
2026/07/04 14:09:30 sca: INFO: Starting evaluation of policy: 'C:\Program Files (x86)\ossec-agent\ruleset\sca\cis_win11_enterprise.yml'
2026/07/04 14:09:30 wazuh-modulesd:syscollector: INFO: Module started.
2026/07/04 14:09:30 wazuh-modulesd:syscollector: INFO: Starting evaluation.
2026/07/04 14:09:30 wazuh-agent: INFO: (4102): Connected to the server ([10.86.27.132]:1514/tcp).
2026/07/04 14:09:30 wazuh-modulesd:osquery: INFO: Module disabled. Exiting...
2026/07/04 14:09:30 wazuh-agent: INFO: (6000): Starting daemon...
2026/07/04 14:09:30 wazuh-agent: INFO: (6010): File integrity monitoring scan frequency: 43200 seconds
2026/07/04 14:09:30 wazuh-agent: INFO: (6008): File integrity monitoring scan started.
2026/07/04 14:09:30 rootcheck: INFO: Starting rootcheck scan.
2026/07/04 14:09:30 wazuh-agent: INFO: Started (pid: 15760).
2026/07/04 14:09:35 rootcheck: INFO: Ending rootcheck scan.
2026/07/04 14:09:36 wazuh-modulesd:syscollector: INFO: Evaluation finished.
2026/07/04 14:10:06 sca: INFO: Evaluation finished for policy 'C:\Program Files (x86)\ossec-agent\ruleset\sca\cis_win11_enterprise.yml'
2026/07/04 14:10:06 sca: INFO: Security Configuration Assessment scan finished. Duration: 36 seconds.
2026/07/04 14:10:37 wazuh-agent: INFO: (6009): File integrity monitoring scan ended.
2026/07/04 14:10:37 wazuh-agent: INFO: FIM sync module started.
2026/07/04 14:10:39 wazuh-agent: INFO: (6012): Real-time file integrity monitoring started.
PS C:\WINDOWS\system32>
```

Connected to the server ([10.86.27.132]:1514/tcp)

• Registering the Windows Agent

Procedure

After updating the manager IP address, the Windows Wazuh Agent successfully established communication with the Wazuh Manager. The connection was verified through the Wazuh Agent log file, where the message "**Connected to the server**" confirmed successful registration.

The Windows endpoint also appeared as **Active** in the Wazuh Dashboard.

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Verification was performed using:

```
sudo /var/ossec/bin/agent_control -lc
```

The output confirmed that the Windows agent was active.

Registered Windows Agent

```
[wazuh-user@wazuh-server ~]$ sudo /var/ossec/bin/agent_control -lc
Wazuh agent_control. List of available agents:
  ID: 000, Name: wazuh-server (server), IP: 127.0.0.1, Active/Local
  ID: 002, Name: Ahmad, IP: any, Active
[wazuh-user@wazuh-server ~]$
```

Windows Agent Visible in Wazuh Dashboard

Agents (1)

⊕

Deploy new agent

🔄

Refresh

📄

Export formatted

More ▾

⚙️

status=active

WQL

☐	ID ↑	Name	IP address	Group(s)	Operating system	Cluster node	Version	Status	Actions
☐	002	Ahmad	10.86.27.102	default	🖥️ Microsoft Windows 11 Pro 10.0.26200.8737	node01	v4.14.6	● active ⓘ	👁️ ⋮

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- **Installation of Wazuh Agent on Kali Linux**

The Wazuh Agent package for Debian-based Linux distributions was downloaded using the **wget** command and installed using the **dpkg** package manager. During installation, the Wazuh Manager IP address was specified so the agent could communicate with the SIEM server.

The Wazuh Agent service was then enabled and started.

Agents (3)									
Search									WQL
☐	ID ↑	Name	IP address	Group(s)	Operating system	Cluster node	Version	Status	Actions
☐	001	Win11	10.86.27.102		-	-	-	● never connected ⓘ	👁️ ⋮
☐	002	Ahmad	10.86.27.102	default	🖥️ Microsoft Windows 11 Pro 10.0.26200.8737	node01	v4.14.6	● active ⓘ	👁️ ⋮
☐	003	kali	10.0.2.16	default	🐧 Kali GNU/Linux 2026.1	node01	v4.14.6	● active ⓘ	👁️ ⋮
Rows per page: 10 ▾									< 1 >

● Agent Registration Verification

The Kali Linux agent successfully connected to the Wazuh Manager. Verification was performed using the Wazuh Manager command:

```
sudo /var/ossec/bin/agent_control -lc
```

```
[wazuh-user@wazuh-server ~]# sudo /var/ossec/bin/agent_control -lc
Wazuh agent_control. List of available agents:
  ID: 000, Name: wazuh-server (server), IP: 127.0.0.1, Active/Local
  ID: 002, Name: Ahmad, IP: any, Active
  ID: 003, Name: kali, IP: any, Active
[wazuh-user@wazuh-server ~]#
```

● Configuring File Integrity Monitoring (FIM)

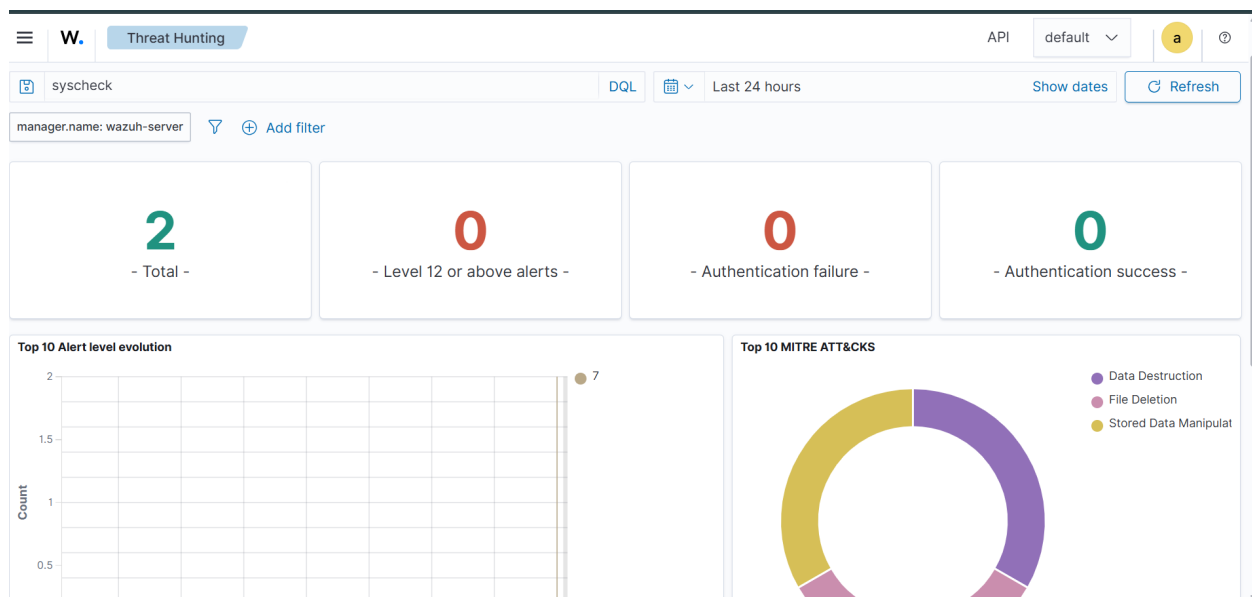
To validate File Integrity Monitoring, a custom folder named **FIM-Test** was created on the Windows desktop. The Wazuh Agent configuration file (`ossec.conf`) was modified to include this directory under the `<syscheck>` section with real-time monitoring enabled.

After saving the configuration, the Windows Wazuh Agent service was restarted to apply the changes.

- **Threat Hunting Dashboard Showing File Integrity Monitoring (FIM) Alerts**

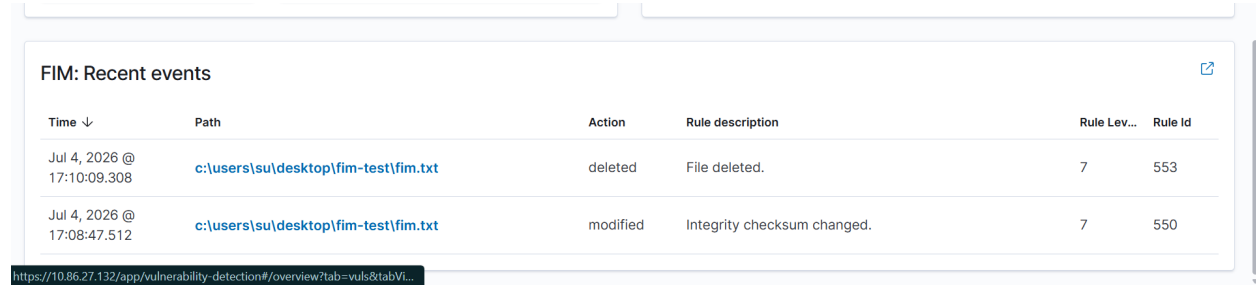
A file named **FIM.txt** was created inside the monitored folder. The file was then modified by adding new content and finally deleted. These actions generated File Integrity Monitoring events, which were successfully detected by the Wazuh Manager.

The generated events appeared in the **Threat Hunting** section of the Wazuh Dashboard as **Syscheck alerts**, confirming that File Integrity Monitoring was functioning correctly



• Security Event Verification

The generated File Integrity Monitoring events were reviewed through the Wazuh Dashboard. The dashboard successfully displayed the alerts generated during the creation, modification, and deletion of the monitored file, confirming successful communication between the Windows endpoint and the Wazuh Manager.



Time ↓	Path	Action	Rule description	Rule Lev...	Rule Id
Jul 4, 2026 @ 17:10:09.308	c:\users\sul\desktop\fim-test\fim.txt	deleted	File deleted.	7	553
Jul 4, 2026 @ 17:08:47.512	c:\users\sul\desktop\fim-test\fim.txt	modified	Integrity checksum changed.	7	550

<https://10.86.27.132/app/vulnerability-detection#/overview?tab=vuls&tabVi...>

• Conclusion

The Wazuh SIEM environment was successfully deployed and configured using the Wazuh Virtual Appliance. Windows 11 and Kali Linux endpoints were successfully connected to the Wazuh Manager through Wazuh Agents. File Integrity Monitoring (FIM) was configured and validated by monitoring changes made to a test file. The generated security alerts were successfully displayed in the Wazuh Dashboard, demonstrating that the SIEM environment was operating correctly. This task provided practical experience in deploying a centralized security monitoring solution and understanding the basic workflow of a Security Operations Center (SOC).